



IDX G9 PHYSICS S STUDY GUIDE ISSUE S2

FINALS

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This guide is a combined guide. Due to the final range including content from before the monthlies, we have included content from the previous issue in this guide for your convenience.

Part 1 Momentum and Impulse

1. Momentum

Definition:

- Momentum (p) is the product of an object's mass (m) and its velocity (v).
- Units: $\text{kg}\cdot\text{m/s}$
- Vector Quantity: Direction is the same as the velocity.

Key Points:

- A truck and a roller skate can have the same momentum if the skate has a much higher velocity.
- Momentum depends on both mass and velocity.
- Mass and velocity control collisions.

Momentum Changes:

- If velocity doubles, momentum doubles.
- If mass doubles, momentum doubles.
- If both mass and velocity double, momentum quadruples.

2. Impulse (I)

Definition:

- Impulse is the product of force (F) and the time (Δt) over which it acts.
- Units: $\text{N}\cdot\text{s}$
- Vector Quantity: Direction is the same as the force.
- Graphical Representation: The area under a Force vs. Time graph equals impulse.

Impulse-Momentum Theorem:

Means that $\Delta p = I$

Key Points:

- Large $\Delta t \Rightarrow$ Small F (e.g., airbags increase collision time to reduce force).
- Small $\Delta t \Rightarrow$ Large F (e.g., hitting a concrete wall stops a car quickly, exerting a huge force).

3. Applications and Problem-Solving Tips

(1) Collisions: Objects with different masses experience the “same force” during collision (Newton’s 3rd Law), but their “momentum changes” depend on mass and velocity.

(2) Graphical Analysis:

(F-t graph)

- Positive area: Impulse in + direction.
- Negative area: Impulse in - direction.

Key equation final review

1. Momentum:

$$\vec{p} = m\vec{v}$$

2. Change in Momentum:

$$\Delta\vec{p} = m(\vec{v}_f - \vec{v}_i)$$

3. Impulse-Momentum Theorem:

$$\vec{F} \cdot \Delta t = \Delta\vec{p}$$

4. Impulse from Graph:

$$I = \text{Area under } F \text{ vs. } t \text{ graph}$$

Important

- Momentum is conserved in isolated systems (no external forces).
- Impulse explains how forces affect motion over time.
- Practice: Solve problems with varying mass, velocity, and force to master these concepts.

Part 2: Conversation of Momentum

1. Definition

Law of Conservation of Momentum: The momentum of any closed, isolated system does not change.

- a. **Closed system:** A system that does not gain or lose mass
- b. **Isolated system:** A system where the net external force on a closed system is zero

Important

-The sum of the momenta of the balls is the same before and after the collision in a closed and isolated system

Formula:

$$\vec{p}_{Ci} + \vec{p}_{Di} = \vec{p}_{Cf} + \vec{p}_{Df}$$

2. Recoil

$$v_{Cf} = \left(\frac{-m_D}{m_C} \right) v_{Df}$$

Introduction to Work

- **Definition:** Work is the use of force to move an object some distance.

Unit of Work: Joule (J)

Scalar quantity. No direction, only magnitude (positive/negative)

Work can be a positive or negative depending on the direction of applied force in relation to displacement of the object.

When the force is a constant and is in the same direction of the motion..... $W = Fd$

At least part of the force must be parallel to the object's displacement $W = Fd \times \cos(\theta)$

θ : the angle between the directions Δd of the force and the displacement

Part 1 Introduction to Energy

- **Definition:** Energy is the ability of an object to produce a change in itself or the world around it.
- **Kinetic Energy (KE):** The energy an object has due to its motion.
- Depends on mass and speed-faster or heavier objects have more KE.
- **Measured in Joules (J).**

$$KE = \frac{1}{2}mv^2$$

- Units: Joule (J), where $1J = 1kg \cdot m^2/s^2$.
- Scalar quantity (no direction).

Part 2 Solving Kinetic Energy Problems

(what can you do to solve these problems)

(1). Finding Speed from Kinetic Energy:

- Given mass and KE, rearrange the KE formula to solve for speed.

$$KE = \frac{1}{2}mv^2$$

- Units: Joule (J), where $1J = 1kg \cdot m^2/s^2$.
- Scalar quantity (no direction).

- Tip: Take the square root at the end to find the final speed.

Important: Always remember to check the units from the given

(2). Finding Mass from Kinetic Energy:

- Given speed and KE, rearrange the KE formula to solve for mass.
- Tip: Ensure units are consistent (kg for mass, m/s for speed).

Part 3 Work Energy Theorem

Definitions

- **Work (W):** Transfer of energy when a force moves an object.

- If force and motion are in the same direction, work is positive (energy increases).
- If force opposes motion (e.g braking), work is negative (energy decreases).
- If force is perpendicular to motion (e.g., circular motion), no work is done.
- **Work-Energy Theorem:**

$$W = \Delta KE = KE_f - KE_i$$

• **Work-Energy Theorem:**

- The work done on an object equals its change in kinetic energy.
- Helps determine final speed, braking distance, or required force.

Part 4 Finding Final Speed After Work is Done:

- Identify initial speed, applied force, and distance moved.
- **Work-Energy Theorem:**

$$W = \Delta KE = KE_f - KE_i$$

- Use the theorem to relate work to the change in KE.
- Solve for the final speed.